

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Shipped 10+ full-stack AI features end-to-end in TypeScript/Node.js: MCP multi-model pipelines (Claude, GPT-4o, Gemini), Temporal.io workflows, real-time event pipelines. - Owned backend stack: PostgreSQL, Redis, Auth0 RBAC, Temporal.io, Docker, Terraform, Sentry; 394 commits across 7 production deployments.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed cloud-based genomic data platform on Google Cloud Run (GCP) with JBrowse 2 genome browser; 100+ researchers, 500GB datasets. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation using OCR, NLP, and ML classification pipelines; improved processing accuracy by 25%. - Automated deployments on AWS and GCP; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

AlgoPulse	Agentic AI Platform	2025
- Agentic AI orchestration on Temporal.io and Kafka; sustained 2,000 async inference req/sec across multi-step reasoning pipelines. - Replaced API polling with Supabase Edge Functions streaming AI reasoning loops; no state loss under load.		
PulseRank	Real-Time Leaderboard	2026
- TypeScript/NestJS leaderboard: MySQL outbox, MongoDB TTL, Redis sorted sets; Testcontainers integration tests. - Shipped OpenTelemetry + Prometheus on all hot paths; load-tested with k6; deployed via GitHub Actions CI/CD.		
LLM Eval Harness	Independent Project	2025
- Vendor-neutral benchmark comparing Claude, GPT-4o, Gemini across 2,000 examples and 3 task types in Python. - Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap + McNemar tests across 6 metrics.		
Glance AI	Privacy-First RAG	2025
- Designed per-tenant RAG for healthcare on GCP with strict vector isolation preventing cross-tenant data leakage and LLM hallucinations. - Deterministic retrieval architecture for regulated environments; factual LLM outputs, no context bleed.		

Languages and Technologies

- Languages:** TypeScript, JavaScript, Python, Java, SQL, Bash
- Fullstack:** React, Node.js, NestJS, Next.js, PostgreSQL, Redis, MongoDB, REST APIs, Docker
- AI/ML & Agentic:** Claude/GPT-4o/Gemini APIs, MCP, LangChain, RAG, LLM evaluation, prompt engineering
- Cloud & DevOps:** AWS, GCP, Terraform, GitHub Actions, Grafana, OpenTelemetry, Kafka, Temporal.io

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog